Assignment 2

RAID(Redundant Array of Independent Disks) Configurations Types of RAID:

Task 1.

RAID Level	Minimum Disks	Data Redundancy	Speed	Real-World Use Case
RAID 0	2	No redundancy; one disk failure causes data loss.	Very Fast	YouTube video editing servers requiring high-speed processing.
RAID 1	2	High; data is mirrored on both disks.	Fast Read, Normal Write	Banking systems storing important customer records.
RAID 5	3	Yes; can survive one disk failure.	Fast Read, Moderate Write	Instagram-like photo storage servers requiring storage efficiency and protection.
RAID 10	4	High; combines mirroring and striping.	Very Fast Read and Write	E-commerce database servers handling high traffic, such as Amazon-like shopping platforms.

Task 2.

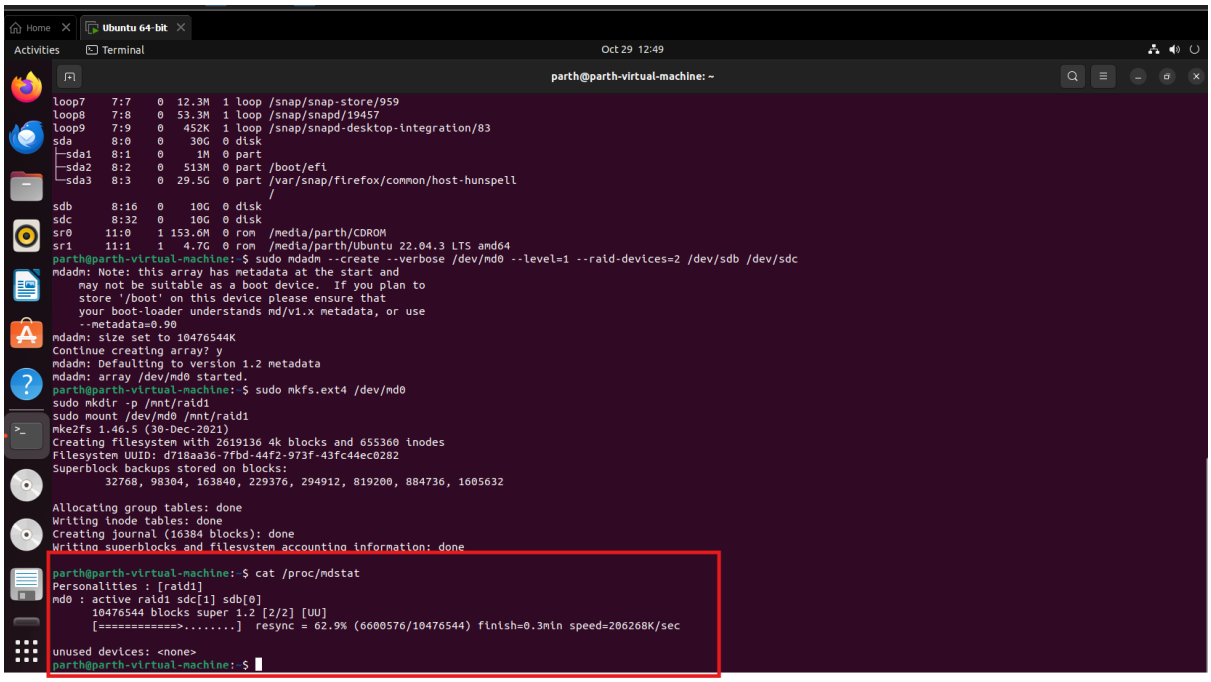
- ## The Setup (Real-Time Cloning):

  - You have two disks: Disk A and Disk B.
  - When WhatsApp saves your chat backup, the RAID 1 system writes the exact same backup file to both disks at the same millisecond.
  - To you, it looks like one single folder, but your data exists in two physical places simultaneously.

- ## What happens if one disk fails?:

  - **Zero Data Loss:** If Disk A suddenly burns out and dies, 100% of your chats and photos remain safe because Disk B holds an identical copy.
  - **No Downtime:** Your server keeps running without crashing. You can still access, read, or download your backups seamlessly from the surviving Disk B.
  - **Easy Recovery:** You simply unplug the dead Disk A, plug in a new empty disk, and the system automatically copies everything from Disk B to restore the twin setup.

Task 3.


Home
Ubuntu 64-bit
Activities
Terminal
Oct 29 12:49
parth@parth-virtual-machine: ~
loop7 7:7 0 12.3M 1 loop /snap/snap-store/959
loop8 7:8 0 53.3M 1 loop /snap/snapd/19457
loop9 7:9 0 452K 1 loop /snap/snapd-desktop-integration/83
sda 8:0 0 30G 0 disk
├─sda1 8:1 0 1M 0 part
├─sda2 8:2 0 513M 0 part /boot/efi
└─sda3 8:3 0 29.5G 0 part /var/snap/firefox/common/host-hunspell
/
sdb 8:16 0 10G 0 disk
sdc 8:32 0 10G 0 disk
sr0 11:0 1 153.6M 0 rom /media/parth/CDROM
sr1 11:1 1 4.7G 0 rom /media/parth/Ubuntu 22.04.3 LTS amd64
parth@parth-virtual-machine:~$ sudo mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 /dev/sdb /dev/sdc
mdadm: Note: this array has metadata at the start and
may not be suitable as a boot device. If you plan to
store '/boot' on this device please ensure that
your boot-loader understands md/v1.x metadata, or use
--metadata=0.90
mdadm: size set to 10476544K
Continue creating array? y
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
parth@parth-virtual-machine:~$ sudo mkfs.ext4 /dev/md0
sudo mkdir -p /mnt/raid1
sudo mount /dev/md0 /mnt/raid1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 2619136 4k blocks and 655360 inodes
Filesystem UUID: d718aa36-7fbd-44f2-973f-43fc44ec0282
Superblock backups stored on blocks:
32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632
Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
parth@parth-virtual-machine:~$ cat /proc/mdstat
Personalities : [raid1]
md0 : active raid1 sdc[1] sdb[0]
10476544 blocks super 1.2 [2/2] [UU]
[============>.....] resync = 62.9% (6600576/10476544) finish=0.3min speed=206268K/sec
unused devices: <none>
parth@parth-virtual-machine:~$

Task 4.

1. **Data Protection:** RAID 5 uses parity information to protect data and can handle one disk failure, while RAID 10 combines mirroring and striping, allowing multiple disk failures as long as both disks in a mirrored pair do not fail.
2. **Performance:** RAID 5 provides good read speeds but slower write speeds because of parity calculations. RAID 10 offers faster read and write performance, making it better for cloud storage services with heavy workloads.

3. **Storage Efficiency and Cost:** RAID 5 uses storage space more efficiently and is cheaper because it requires less redundancy. RAID 10 uses half of its total disk capacity for mirrored copies, making it more expensive but generally faster and more reliable.